# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – SHORT ANSWER TEST

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|---|---|---|---|
| Course Code: | CSA16 | Course Title: | Data Warehousing and Data Mining |
| CO Assessed: | CO1 – Precision (BL1, BL2) | Assessment: | Assessment Tool 1 – Short Answer Test |
| Weightage: | 10% of CIA | Total Marks: | 100 (5 Questions × 20 Marks) |
| CO1 Statement: | *Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2* | | |
| Student Name: | N.Mokshagna | Reg. No.: | 192472205 |
| Section / Batch: | CSE-A1 / 2024 | Date: | 15/07/26 |

**Instructions to Students**

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

| Question Type | Focus Area | Questions | Marks |
|---|---|---|---|
| Short Answer Test | Architecture of Data Warehouse, ETL, OLAP | Q1 – Q5 | Q1 –20<br>Q2 –20<br>Q3 –20<br>Q4 –20<br>Q5 –20 |
| | | GRAND TOTAL: | 100 Marks |

## SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.
(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.
(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
  - Product: Category, type, and batch.
  - Factory: Location, department, and line number.
  - Time: Year, quarter, month, and week.
  - Supplier: Region, rating, and contract type.
- Measures:
  - Defect rate.
  - Production units.
  - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
  - Product: Category, subcategory, and SKU.
  - Store: Region, city, and store ID.
  - Time: Year, quarter, month, and day.
  - Customer: Membership tier, age group, and gender.
- Measures:
  - Inventory turnover.
  - Total sales revenue.
  - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

**Code of Conduct**

*I N.Mokshagna / 192472205 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: N Moksha

## RUBRICS FOR CALCULATION

| Criteria | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | Demonstrates thorough, accurate understanding of concepts | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation | 25% | Detailed, well-explained answers with relevant examples | Clear explanation with adequate details | Limited explanation; lacks depth | Poor or unclear explanation |
| Application | 20% | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors | Limited application with weak analysis | No application or incorrect analysis |
| Organization | 15% | Well-structured answers with logical flow and coherence | Organized with minor issues in flow | Basic structure, lacks clarity | Poor organization, incoherent answers |
| Accuracy | 10% | Completely accurate and covers all required points | Mostly accurate with minor omissions | Partially correct with missing points | Incorrect answers with major gaps |
| Clarity | 5% | Clear, neat, professional presentation | Understandable with minor issues | Limited clarity, readability issues | Poorly written, difficult to understand |

## Marks Evaluation:

| Question No. | Understanding of Concepts (25%) | Explanation (25%) | Application (20%) | Organization (15%) | Accuracy (10%) | Clarity (5%) | Total Marks |
|---|---|---|---|---|---|---|---|
| Q1 (20 Marks) | | | | | | | |
| Q2 (20 Marks) | | | | | | | |
| Q3 (20 Marks) | | | | | | | |
| Q4 (20 Marks) | | | | | | | |
| Q5 (20 Marks | | | | | | | |
| | | | | | Overall Total (100) | | |

**Faculty In-charge**

Signature: ________________

Date: ________________

**Course Coordinator**

Signature: ________________

Date: ________________

**Head of Department**

Signature: ________________

Date: ________________

N.Mokshagn
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1) City Hospital Data warehouse.

a) Star schema

Fact table:- Consulation - Fact

* measures : count : Change, avg - Charge.

Dimension Tables:-

* patient - Dim (ID, name, Age, Gender)
* Doctor - Dim (ID, name, specilization)
* Department - Dim (ID, name).
* Time - Dim (Day, month, quarter, year)

b) OLAP Operations:-

* Roll-up : time → year
* slice : select Average
* Dice : Department -wise data

c) no:- of cuboids

formula:-

$(n+1)^d$

$= (4+1)^4 = 625$ cuboids

2) smart city project:-

Star schema

Fact table:- smart city - Fact

measures:-

* Vehicle count
* Congestion level
* Punctuality
* Air quality index (AQI)

Dimensions:-

* Time
* location
* Transport
* Sensor

(3) Banking Data warehouse

a) Galaxy schema

Fact table:-

* Transaction - Fact (Amount, count)
* Loan - Fact (Amount, interest)

Shared Dimensions:-

* Customer
* Branch
* Time
* Account type

b) OLAP Operations:-

* Roll-up : Time -> Year
* Dice : Branch-wise data

measure:- Average Transaction Amount and Loan Amount

c) cuboids

$(4+1)^4 = 625$ cuboids

4) Manufacturing company

a) Roll-up

* summarize Defect Rate by product category across all factories for the current year.

b) Drill-down

* Analyze Defect rate by factory → Department → product type.

c) slice.

select supplies with rating = 4+ across all regions

d) Dice

* Analyze Defect rate and supplier reliability for selected product categories factory locations.

5) Retail chain

a) Roll-up

summarize inventory Turnover by product category across all stores for the last quarter

b) Drill - down

=) Analyze electronics sales: Region → sub category → SKU.

c) Slice

View total sales revenue for male customers with Bronze membership.

d) Dice:-

Analyze sales revenue and units sold for selected cities, product categories and time periods.